Risk-Aware Parameter Optimization of Additive Manufacturing Process via Conditional Diffusion Model with Co-Registered Digital-Twin Data

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Abstract

Product quality and reliability in additive manufacturing vary sharply with changes in process parameters such as hatch spacing, laser power, and scan speed. For any given parameter combination, qualification metrics like volumetric defect rate follow a probabilistic distribution. Thus, a "best" parameter set may minimize the mean defect rate yet still expose the process to heavy-tail risk. Because each production run consumes substantial time and material, manufacturers must both identify risk-aware parameter sets and make real-time, *in-situ* decisions during a build to minimize and correct for potential defects. Leveraging recent advances in generative modeling and predictive machine learning, we propose a data-driven framework that forecasts future-layer defect rates and supports risk-aware, *in-situ* decision-making to reduce scrap, rework, and cost.

We adopt a unified *generator–evaluator–decision* framework trained across observation ratios of $r \in \{0.1, 0.2, \ldots, 0.9\}$, connecting monitored layer history to risk-aware control settings during printing. A conditional diffusion *generator* learns $p_\theta\big(X_{\text{tar}} \mid X_{\text{obs}}, s\big)$, stochastically extrapolating unobserved *future* layers X_{tar} from partial observations X_{obs} under a candidate processing-parameter schedule 's' for each ratio 'r'. For each (s, r), we draw samples $\{X_{\text{tar}}^{(m)}\}_{m=1}^{M}$ that preserve co-registered geometry and scan-path structure while reflecting process variability. The *evaluator* maps each synthesized signature to a scalar defect metric $Y^{(m)}$ (e.g., porosity rate or defect segmentation rate), yielding an empirical distribution $\{Y^{(m)}\}$ from which we compute the mean loss $\mu(s, r)$ and tail risk $\text{CVaR}_\alpha(s, r)$. Using these quantities, we select a risk-aware parameter set such that

$$s(r) = \arg\min_{s} \big[\mu(s, r) + \lambda\, \text{CVaR}_\alpha(s, r)\big].$$

With the selected set of processing parameters s, a lightweight ML(machine learning)-based *decision* model predicts $\hat{Y} = f\big(X_{\text{tar}}, s\big)$ to trigger GO/NO-GO alerts and process parameter updates via a calibrated threshold τ, closing the loop from partial observations to risk-aware, in-process quality control.

Our framework is evaluated on the Peregrine co-registered digital twin dataset (v.2023-11), a hierarchical HDF5 file that aligns layer-wise visible images, scan path vectors, and synchronized process logs with relevant component and sample indices. For this study, we train models primarily on Build 1 (3,575 layers, 39 parts, 3,296 SS-J3 sample coupons of 316L) and use Builds 2–5 to assess cross-platform, multi-laser, and process-contrast generalization. This dataset ensures that X_{obs} and X_{tar} share the same coordinate frame, allowing the generator to produce physically consistent completions and the evaluator to compute defect rates that are comparable across settings s and observation ratios r, thereby directly supporting the risk-aware selection of $s(r)$ as well as a reliable real-time decision maker.

Anticipated contributions are twofold: (i) from a manufacturing perspective, we demonstrate a risk-aware parameter selection and *in-situ* control by selecting $s(r)$ to jointly lower the mean defect $\mu(s, r)$ and tail risk $\text{CVaR}_\alpha(s, r)$ while improving GO/NO-GO reliability from early-layer evidence; and (ii) from a modeling perspective, we provide a conditional diffusion model trained on a multimodal digital twin under varying observation ratios, and report ablations versus anomaly prevalence to enable quantitative and physically consistent analysis.

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